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Section A

Ans1 CPU register are small, high speed storage locations inside the processors.

Imp:
Store intermediate data and Instruction
Fasters than cache and memory.
Improve execution speed of program

Ans2 Program Counter is a register that holds the address of the next instruction.

Ans3 Instruction format defines the structure of an instruction in binary form.
Key Fields:-
Opcode (operation)
Operand
Addressing mode

Ans5 Addressing mode specify how to locate operands
Purpose:-
Provide Flexibility in accessing data.
Reduce instruction count
Improve program efficiency

Ans6 ISA is the interface between hardware and software defining supported instructions.
Includes.
Instruction Formate
Data types
Registers and addressing mode.

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Ans 7 Types of instructions in a processor are:
Arthemetic (ADD, Sub)
Logical (AND, OR)
Data Transfer (LOAD, STORE)
Control (JUMP, BREAK)

Ans 8 Explanation
Opcod is interpreted
Required operation is identified
Control signals are generated
Operands are prepared for execution

Ans 9. Basic instruction cycle are
Fetch instruction from memory
Decode instruction
execution operation
Store result

Ans 10. Characteristics of RISC processors are:
simple and fixed length instructions.
Few addressing mode
Faster execution (pipeline - Friendly)
Load/store architecture.

Ans 11. Characteristics of CISC processor are:
~~Simple~~ Complex and variable length instructions
Many addressing per program
Slower but powerful instructions.

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Ans 12: RISC-V:

Open Source ISA

Simple, fixed-length instructions

Easies to design and extend.

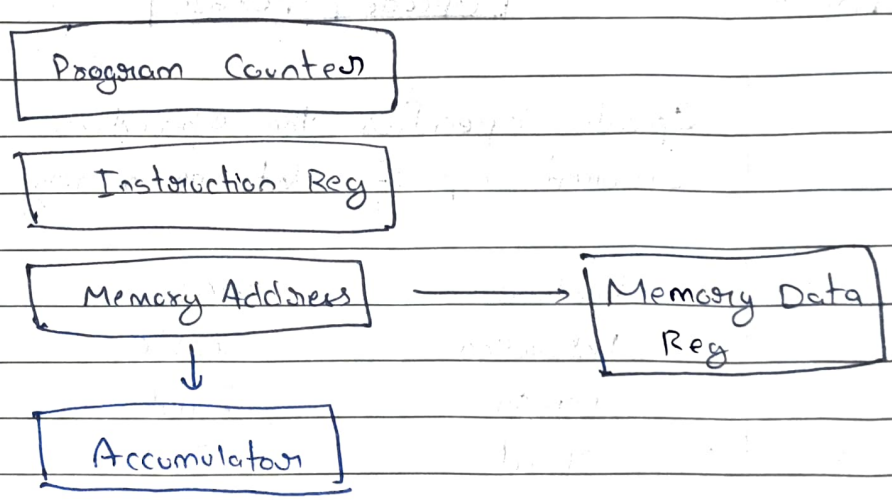
X86:-

↳ Proprietary (Intel / AMD)

↳ Complex variable-length instructions.

Section B

Ans 1



PC holds next instruction address, IR stores current instruction.

MAR & MDR handle memory access: Acc stores intermediate results

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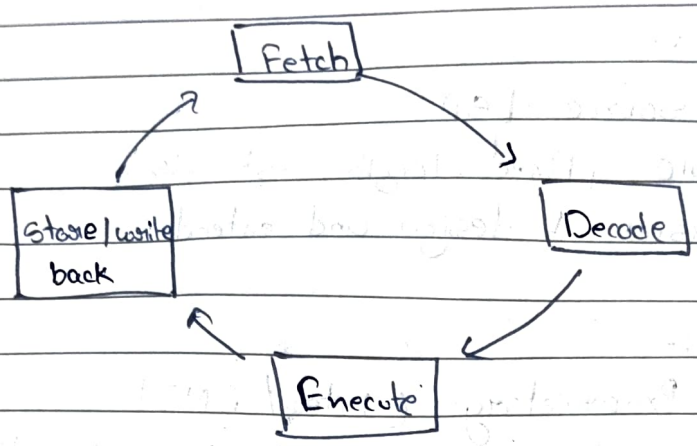
Ans 2



CPU fetches instructions, decode it, execute it and write back

Ans 3

Opcode	operands

Opcode specifies the operations.
operand specify data or memory location

Feature	RISC	CISC
Instruction	Simple	Complex
Length	Fixed	variable
Speed	Fast	slower
Memory	More usage	less usage

RISC focuses on simple and speed
CISC focuses on powerful and complexity

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Section → C

Ans1 Impact on performance are:-
Memory access is slower than register access
Increases CPI due to memory latency
Overall execution time increases → Performance decrease

Ans 2 The should choose RISC architecture because:-
Simple and fixed-length instructions
Easier to pipeline efficiently
Requires simple hardware design.

Ans 3 CPU executes then instruction as:
Step 1 : Fetch
Instruction is fetches from memory into IR

Step 2: Decode
Opcode is decoded and requires are identified

Step 3: execution
Data is read from registers
ALU perform operation
Result is stored back in ~~the~~ register.

Ans 4 X86 is widely used despite complexity because:
Backward compatibility with older software.
Large ecosystem (OS, app, industry, support)
Modern CPU internally optimized

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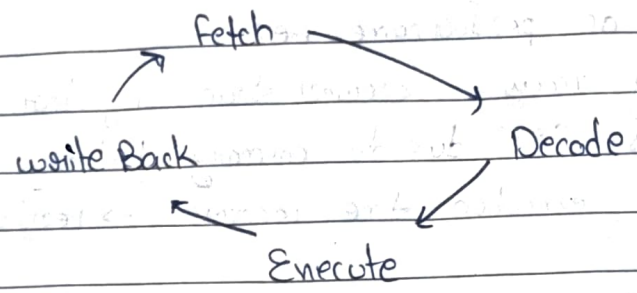
Section - D

Ans 1



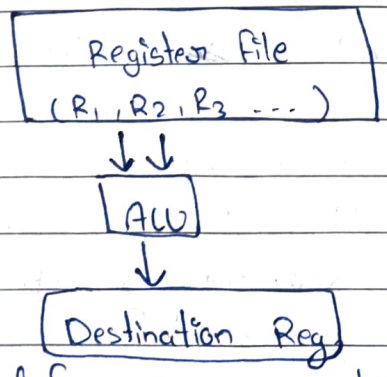
PC holds address of next instruction
Address sent to memory via MAR
Instruction fetched into MDR and loaded onto IR
PC is incremented to next instruction.

Ans 2

Funct 7	rs2	rs1	Funct 3	rd	opcode

opcode - specifies operation type
rd - Destination register
rs2, rs1 - Source register
Funct 3, 7 → Define exact operation
Fixed Length (32-bit) format simple decoding

Ans 3:



Data is read from source register
Sent to ALU for operation
Result is written back to distination register
Enable fast computation without memory access.